

‘A Fine Balance’: Exploring the Interplay of Reproductive Strategies and Growth Dynamics in Trees

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Masting is the phenomenon of synchronized and highly variable seed production within plant populations. It plays a critical role in ecosystem-level function, influencing both plant regeneration and the population dynamics of seed predators. Despite its importance, models of it are poorly predictive, likely because of the complex temporal dependencies and the difficulty of integrating environmental variation with internal plant dynamics. One of the most important internal plant dynamics hypothesized to drive masting is a trade-off between vegetative growth (leaves, stems, wood) and reproduction (seeds). However, this trade-off is often obscured by environmental variation at the same time. Here we leverage rarely available coupled data over time and space with a new hierarchical Bayesian generative model to test for evidence of this trade-off while also accounting for variation in the environment.

The model explicitly treats annual resource allocation as a latent process occurring the individual tree level, informed by year-to-year environmental variation and two distinct data sources: annual tree-ring widths as a proxy for growth and annual seed production records as a proxy for reproduction collected from Mount Rainier National Park. Our approach then reconstructs hidden resource pools from observed ecological data and tests how they shift with the environment regarding masting events, specifically modeling shifts in snowpack that shape moisture over the growing season and temperatures that drive growing season warmth. Uniquely our data includes multiple stands across an elevational gradient, allowing us to model how environmental variation both between years and within years across space may shape resources pools and allocation. We utilize hierarchical structures to pool information across individuals, stands and species to help to estimate the latent state transitions and how they vary across scales. This model addresses the use of Bayesian modeling and latent variable frameworks in deciphering unobservable physiological processes from observable ecological data.